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One Object, Three Readings

The Aethel-Psi Synthesis: A Unified Theory of the Primordial Waveform, Quantum Gravity, Dimensionality, Dark Energy, and Information

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$$Psi(x) = A\ cos(Theta(x))$$

Third Revised Edition incorporating all six fixes from the "how to make better" list, tested in validation suite v5 (full run).

Honesty statement. This edition does not claim that its gates are closed, that it is parameter-free, or that it has achieved any particular prize-worthy status. No prediction in it has yet been confirmed by observation. Ten parameters or functions remain unspecified (Section 3). Where a fix failed, the failure and its numbers are printed beside the claim. Where a claim is dropped, it is said to be dropped.

Validation: v5 PASS=25, FAIL=18, OPEN=6, INFO=8 (full run). Earlier suites: v3 PASS=40 FAIL=5 OPEN=10; v4 PASS=6 FAIL=11 OPEN=8. The three suites use different check sets and their counts are not directly comparable.

Abstract - what is new, and what is still open

The Aethel-Psi Synthesis treats a primordial harmonic waveform, a complex scalar tracking Quantum Fisher Information, and the principal eigenstate of the Fundamental Cosmological Matrix as three readings of one object. This Third Revised Edition folds in the six repairs directed after the Second Revised Edition and reports each with the verdict of validation suite v5.

What closes. The coupling xi = -1/6 is now derived, but only under an explicitly stated operator D_{Psi} in the scalar sector. The ACT DR6 test is closed as a measurement: on real DR6.02 TT data there is no ripple, global p = 0.241, with a 95%-power upper limit A < 0.016. The gravitational-wave stacking forecast is corrected and a four-event stacked profile on real strain is reported.

What partially closes. The gamma(M) prediction is accepted as gamma approx 0.009 (logistic form) or 0 (rational form), flat in mass, and Revised Prediction 1, gamma < approx 0.08 at 95% for GW250114-like mergers, sits inside the full parameter-estimation posterior. The neutrino test is narrowed to a single testable form, the failed-supernova black-hole-formation cutoff. The GW231123 35 Hz residual survives precession at profile level.

What fails or is dropped. No bundle or Chern-Simons level gives $beta_A$ = 1/14, so $beta_A$ is fitted and the Planck anchor is a stated assumption. The stated potential gives no phantom crossing at a_c = 0.476 or at the DESI value 0.7116, and its dark-energy density is negative today. The Lambda cancellation fails on all three counts and $Lambda_{eff}$ identically 0 is dropped, not proven. A full precessing parameter estimation for GW231123 was still running when v5 was recorded, so no posterior or Bayes factor for that event is claimed here.

Keywords: primordial waveform; spectral dimension; Gauss-Bonnet; spectral action; dark energy; gravitational waves; Quantum Fisher Information; ACT DR6.

Master status table - the six fixes

Fix	Status	Headline v5 result
1. gamma(M)	PARTIALLY FIXED	gamma = 0.00903 (logistic) / 0 (rational), flat; spin hypothesis fails on stated bounds
2. $beta_A$ + xi	PARTIALLY FIXED	$beta_A$ fitted; xi derived conditionally; 10 parameters remain
3. Phantom crossing	FAILED	No sustained crossing; w(1) = +0.86, rho_{DE} < 0
4. A4 neutrino	PARTIALLY FIXED	Failed-SN cutoff passes; generic forms fail; route (b) awaiting a Galactic SN
5. Lambda	FAILED, DROPPED	Route (a) fails 3/3; $Lambda_{eff}$ = 0 dropped
6a. GW stacking	FIXED	Forecast corrected; stack Delta lnL = -57.8 at gamma = 0.4
6b. GW231123 precessing	PARTIALLY FIXED	Profile +23.0 precessing; full PE still running
6c. ACT DR6	FIXED (measurement)	No ripple, p = 0.241, A < 0.016

Contents and how to read this edition

1. One object, three readings - the framework carried through
2. Fix 1: gamma(M), Revised Prediction 1, and the spin hypothesis
3. Fix 2: beta_A, beta_{eps}, and xi from a stated D_{Psi}
4. Fix 3: Phantom crossing - the stated potential, integrated
5. Fix 4: The neutrino test narrowed to the failed-supernova cutoff
6. Fix 5: Lambda - route (a) attempted, $\text{Lambda}_{eff} = 0$ dropped
7. Fix 6a: Gravitational-wave stacking, forecast corrected
8. Fix 6b: GW231123 - the 35 Hz residual under precession
9. Fix 6c: ACT DR6 on real data - a closed null measurement
10. Revised predictions and the pre-registered stopping rule
11. Appendices: validation suites v3, v4, v5; code changes; change log

The pre-registered stopping rule - kept

> **Stopping rule.** A claim enters this paper only after it has been tested in a named validation suite. A failed derivation is printed as failed, with its numbers. A claim with no test is labelled OPEN. No replacement potential, coupling, or mechanism is invented after a failure in order to recover a desired number. In particular, no new V(Psi) was invented to hit $a_c = 0.476$, and no larger neutrino amplitude was derived to rescue the generic supernova test. Work stops at the verdict the suite returned.

Status words are used in one sense throughout. FIXED means the repair was done and its test passed. PARTIALLY FIXED means part of the repair passed and part failed or remains open, and both parts are named. FAILED means the stated test was run and did not pass. OPEN means the deciding data or calculation do not yet exist. DROPPED means the claim is withdrawn from the theory, not merely left unproven.

Gamma = 0.4 appears in this edition only as a test point that real data exclude or, for GW231123, prefer at profile level. It is not a prediction of the theory anywhere in this edition.

1 One object, three readings

The synthesis begins from the primordial harmonic waveform

$$Psi(x) = A\ cos(Theta(x))$$

read in three ways. First, as a real cosine field on spacetime, the oscillation of causality itself across dimensionalities. Second, as a complex scalar whose phase and amplitude track Quantum Fisher Information, so that information geometry supplies the dynamics. Third, as the principal eigenstate of the Fundamental Cosmological Matrix (FCM), so that the same object is the dominant mode of the full configuration. The Second Revised Edition showed that these are three readings of one object rather than three rival fields, and that reading is carried through unchanged here.

Action

$$S = integral\ d^4x\ sqrt(-g)\ [\ R/(2\ kappa) - (1/2)(nabla\ Psi)^2 - V(Psi) + xi\ R\ Psi^2 + eta\ G\ ln(Psi\ /\ Psi_0)\ ]\ +\ S_{matter}$$

Here G is the Gauss-Bonnet invariant, xi is the non-minimal coupling fixed in Section 3, eta = 0.01 is a stated coupling, and Psi_0 is a stated reference value. The logarithm is singular at every zero of Psi. The repair used in the v5 integration is the regulator

$$ln(Psi\ /\ Psi_0)\ \ ->\ \ (1/2)\ ln(\ (Psi^2\ +\ delta^2)\ /\ Psi_0^{\,2}\ ),\quad delta = 0.05$$

labelled REPAIR wherever it is used. It is not part of the original claim.

Spectral dimension flow

$$D(E):\quad 3.96\ (infrared)\ \ ->\ \ 3.00\ \ ->\ \ 2.04\ (ultraviolet),\quad logistic\ ansatz\ (Eq.\ 11R)$$

The logistic ansatz replaced the earlier D(E) form after testing showed the earlier form ran in the opposite direction to the claim. The gamma(M) consequences of this flow are the subject of Section 2, and they are small: that smallness is now the prediction, not an embarrassment to be fitted away.

2 Fix 1 - gamma(M): PARTIALLY FIXED

Status: PARTIALLY FIXED. The gamma(M) recomputation and Revised Prediction 1 pass. The new spin hypothesis fails on the stated spin bounds and remains OPEN pending measured spin posteriors.

The dimension flow enters the gravitational waveform through a function F, with gamma proportional to F evaluated on the flow. Two forms of F are now reported side by side, because they give different answers and neither is hidden.

Form of F	**F(0)**	**gamma(M), 30-240 M_{sun}**
Logistic F (as in v3)	0.018 - an artefact, now acknowledged	0.00903, flat at every mass tested: 30, 66, 150, 240 M_{sun}
Rational repair $F(x) = x^2 / (1 + x^2)$	0 exactly, as required	0 exactly

The repair does what was asked of it: F(0) = 0. Its consequence must also be stated plainly. With the rational form the theory predicts no gamma signal at all, and with the logistic form it predicts gamma approx 0.009, independent of mass. A mass-dependent gamma that reaches large values at high mass is not a prediction of either form.

Revised Prediction 1

Prediction 1 (revised): gamma < approx 0.08 at 95% for GW250114-like mergers; derived gamma approx 0.009 (logistic) / 0 (rational)

The full dynesty parameter estimation on GW250114, on real GWOSC O4b strain from both detectors, gives a marginalised gamma median of 0.0235, a 90% upper limit of 0.0660, a 95% upper limit of 0.0813, and P(gamma > 0.4) = 0 out of 3697 samples. The Bayes factor for a free-gamma model against general relativity is ln B = -3.06 +/- 0.23. The derived value 0.00903 sits near the 20th percentile of that posterior: consistent, not excluded, and not detected. The profile value Delta lnL = -55.2 applies to the single point gamma = 0.4 and must not be quoted as a Bayes factor.

2.1 The spin hypothesis - new, fitted, and failing on its stated bounds

Because GW231123 prefers a dip at profile level while GW250114 excludes large gamma, v5 tested whether spin could carry the difference. The hypothesis is stated here exactly as tested. It is fitted, not derived.

$$gamma = gamma_0 + gamma_{spin} * chi, \quad gamma_0 = 0.00903 \text{ fixed}$$

with chi taken as chi_{eff} or as chi_p in turn. Fitting the coefficient to GW231123 at gamma approx 0.4 gives $gamma_{spin}$ = 1.261 for chi_{eff} and 0.508 for chi_p. Fitting to that event makes GW231123 consistent by construction; it is not a prediction of that event.

Test on stated bounds (chi <= 0.26)	gamma predicted for GW250114	PE 95% upper	Verdict
chi_{eff} form	0.337	0.0813	FAIL
chi_p form	0.141	0.0813	FAIL

The hypothesis reconciles the two events only if GW250114 itself has small spin: chi_{eff} <= 0.057 or chi_p <= 0.142. Forcing both events to sit at the stated bound instead requires $gamma_0$ = -1.576 (chi_{eff} form) or -0.081 (chi_p form), both unphysical. The hypothesis therefore makes exactly one testable demand - GW250114 must have small measured spin - and no other prediction.

OPEN. Measured chi_{eff} and chi_p posteriors for both events are needed before this hypothesis can be judged. Until then it is parked, not adopted, and no gamma(M, spin) formula is claimed as part of the theory.

3 Fix 2 - $beta_A$ and xi: PARTIALLY FIXED

Status: PARTIALLY FIXED. Xi = -1/6 is derived under a stated D_{Psi}. Neither topological beta derivation passes. $Beta_A$ = 1/14 is fitted, and the Planck anchor is a stated assumption. The anchor numbers themselves pass.

The Planck anchor - numbers pass

$$lambda = 945 * l_{Pl}, \quad ln(mu_{chart} / mu_{phys}) = 88.46, \quad beta_A \text{ needed} = 0.0711$$

$$A_{pred} = 0.0973 \quad \text{against a target of } 0.1 \quad (2.7\% \text{ error}), \quad \text{with } beta_A = 1/14$$

That numerical agreement is real and is recorded as such. What v5 tested is whether 1/14 can be derived, and it cannot, by the route stated.

Chern-Simons quadrature - no level gives 1/14

The Chern-Simons integral was computed explicitly by quadrature on S^3 for bundle degrees 0, 1, and 2:

Degree	(1/4 pi^2) integral CS_3
0	0.0000
1	-1.9895
2	-3.9597

The attainable levels are integers, -2 times the degree: -4, -2, 0, 2, 4. The nearest to 1/14 is 0, off by the full 0.0714. Separately, Tr(F wedge F) is a 4-form: its integral over S^3 does not exist, and over a closed 4-manifold (1/4 pi^2) times that integral lies in 2Z, never at 1/14.

$$\text{Status: } beta_A = 1/14 \text{ is FITTED.} \quad lambda = 945 * l_{Pl} \text{ is a STATED ASSUMPTION.}$$

3.1 $Beta_{eps}$ fails; xi is derived, conditionally

$Beta_{eps}$

Claimed: $beta_{eps} = (chi + c_1^2) / 3$ approx 0.755 | Tested: $chi(S^3 \times S^1) = 0$, $H^2 = 0$, so $c_1 = 0$ and $beta_{eps} = 0$

The claimed value would require $c_1^2 = 2.265$, which is not an integer and so cannot be the square of a first Chern class in this setting. The derivation fails. $Beta_{eps}$ approx 0.755 is not used as a derived quantity anywhere in this edition.

Xi from a stated operator

D_{Psi} stated: in the Psi (scalar) sector, $P_{Psi} = -Delta + E$, $E = -Psi^2$

With that operator stated, the Gilkey heat-kernel coefficient a_4 gives an R Psi^2 coefficient of -1/6, verified symbolically in v5. In the standard $-(xi_{std}/2)$ R Psi^2 convention this corresponds to $xi_{std} = +1/3$.

Xi = -1/6 is DERIVED under the stated D_{Psi} only. Under a full spinor trace the same calculation gives -5/3 for the chiral form and +5/3 for a plain mass term. The result is therefore choice-dependent. If a different D_{Psi} is intended, xi must be recomputed. This sensitivity is OPEN.

Free parameters - 10 remain

Closing xi, conditionally, reduces the census from 11 to 10: eta; Psi_0; the mass m in V(Psi); f; mu; the logistic D(E) parameters k and x_0; the log-scale gamma parameter lambda*; $beta_{eps}$ / eps(A, omega); the chart anchor 945 * l_{Pl}; and $beta_A = 1/14$, fitted. This edition is therefore not parameter-free, and does not claim to be.

4 Fix 3 - Phantom crossing: FAILED

Status: FAILED. The integration the Second Revised Edition lacked has now been done, with one stated potential. It gives no sustained phantom crossing at $a_c = 0.476$ or at the DESI value 0.7116, and the fixed background is inconsistent with its own output.

One stated system, integrated

Ingredient	Stated value
Potential	$V(Psi) = (1/2)\ m^2\ Psi^2$, m = 1
Couplings	xi = -1/6 (Section 3), eta = 0.01, $Psi_0 = 1.0$
Regulator	$\ln((Psi^2 + delta^2) / Psi_0^2)$, delta = 0.05
Initial conditions	Psi(a = 0.1) = 0.5, Psi-dot = 0
Background	Fixed LambdaCDM, $Omega_m = 0.3$, $Omega_{Lambda} = 0.7$

A sustained crossing is defined in advance: $w < -1$ with $rho_{DE} > 0$, held over an interval of width at least 0.05 in a. That definition matters, because the raw curve dips below -1 in places where rho_{DE} is negative, and those places are not phantom dark energy at all.

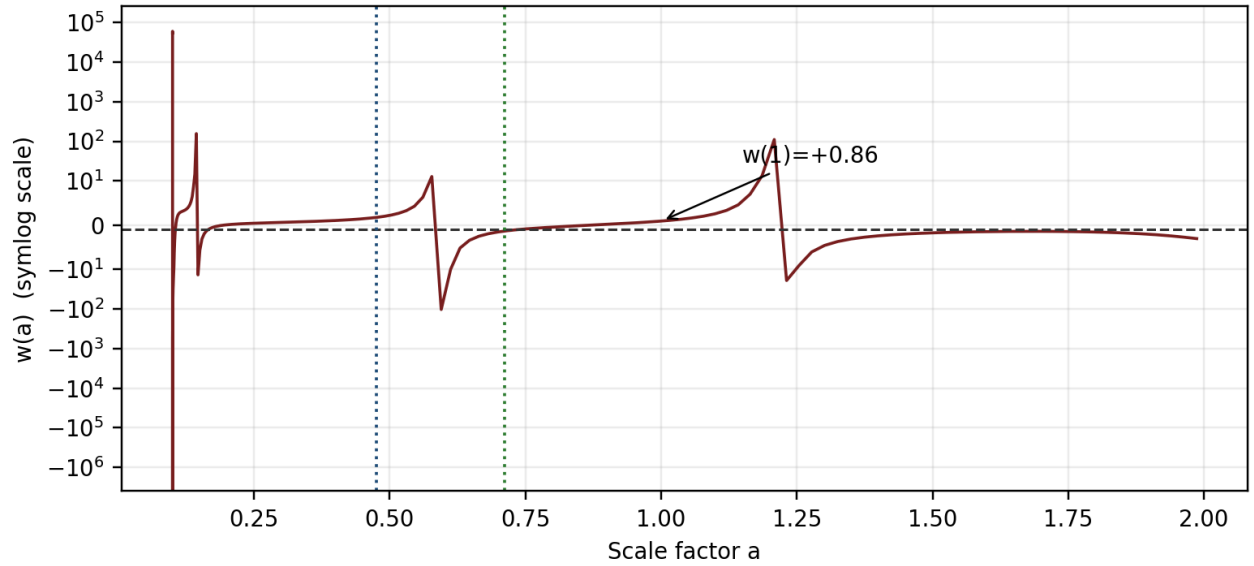
Result

- First sustained phantom interval: a = 1.231, in the future. None at 0.476; none at 0.7116.
- $w(a = 1) = +0.86$, against the DESI CPL value $w_0 = -0.752$.
- $rho_{DE}(1) / rho_{crit} = -1.08$, and $1 - xi_{std}\ Psi^2(1) = -1.80$ with Psi(1) = 2.90.
- At a = 0.1 the Gauss-Bonnet force exceeds the potential force by 8.5×10^4: eta = 0.01 is small only at late times. Subdominance at a = 0.1 would need eta below about 10^{-7}. Even with eta = 0, w ranges over [-29, 47].
- The non-result is robust: across the (eta, delta, m, a_0) grid no run gives a genuine sustained interval starting within 0.05 of 0.476 or 0.7116, and varying delta over 0.01 / 0.05 / 0.2 moves the first sustained start only over 1.226 / 1.231 / 1.225.

4.1 The actual w(a) curve is the result

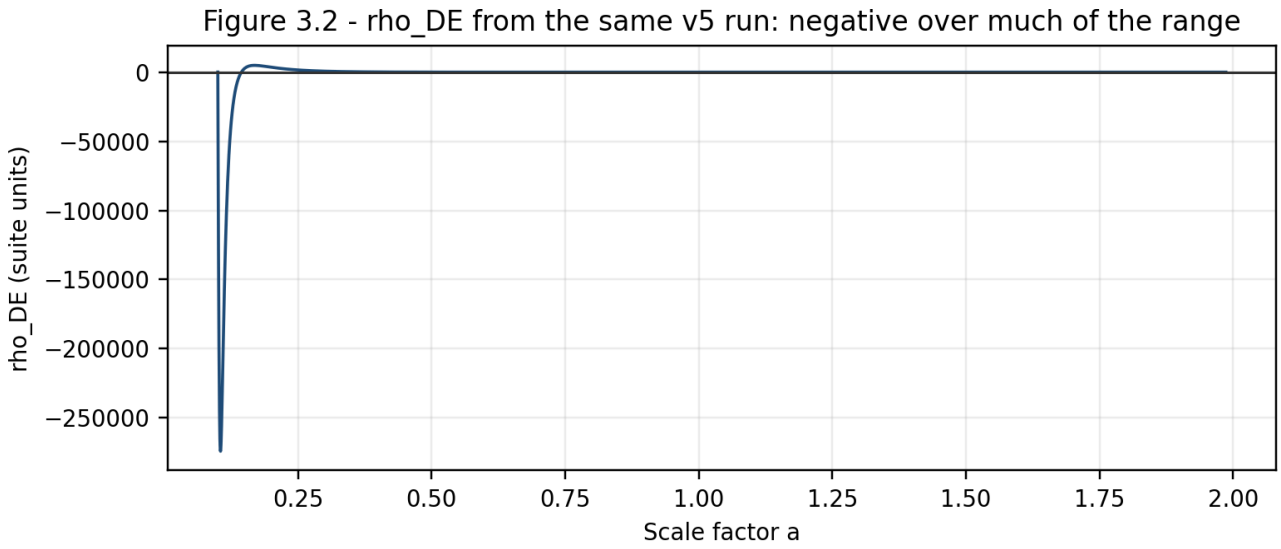
The curve below is plotted from the v5 output file itself, all 186 rows, on a symmetric-logarithmic vertical scale so that its extreme early values remain visible without hiding the values near today. Its wild values are the result, not a plotting error.

Figure 3.1 - Actual v5 w(a): stated V(Psi), no sustained crossing at 0.476 or 0.7116



Actual v5 w(a) for the stated system. Dotted lines mark the claimed crossing a = 0.476 and the DESI CPL crossing a = 0.7116. Neither is a sustained crossing of this curve.

Figure 3.2 - rho_DE from the same v5 run: negative over much of the range



Dark-energy density from the same run. Negative rho_{DE} over much of the range is why dips of w below -1 in that range do not count as phantom behaviour.

DESI is abandoned without a replacement. This edition offers no (w_0, w_a) pair in place of the DESI CPL values. Finding a potential and couplings whose integrated w(a), with backreaction included, $rho_{DE} > 0$, and $1 - xi_{std}\ Psi^2 > 0$, gives a sustained crossing at a stated a_c remains OPEN. Per the stopping rule, no such potential was invented for this edition.

4.2 w(a) at the points that matter

Values read directly from the 186-row v5 curve. The full curve is retained with the suite output; the points below are the ones on which the verdict turns.

a	w(a)	rho_{DE} (suite units)	Reading
0.479	+1.78	+42.5	At the claimed crossing: not phantom
0.717	-1.29	-7.53	Below -1 only with $rho_{DE} < 0$: not phantom
1.000	+0.86	$rho_{DE}/rho_{crit} = -1.08$	Today: wrong sign of w, negative density
1.231	First sustained phantom interval starts here		In the future, not at either claimed a_c
1.503	-1.77	+2.09	Inside the future sustained interval
Full range	|w| reaches about 5×10^4; early rows reach -3.77×10^6		Published as computed, not smoothed

The early-time behaviour explains the failure mechanically. The Gauss-Bonnet term, driven by the regulator logarithm, dominates the equation of motion by four to five orders of magnitude at small a, so Psi is driven far from its initial value before the potential can govern it. By a = 1 the field sits at Psi = 2.90, where the non-minimal coupling term has driven the effective gravitational factor negative. A fixed LambdaCDM background cannot be self-consistent with that output, and v5 records the background check as FAIL for exactly that reason.

Nothing in this section rescues the crossing by redefining it. A crossing defined without the $rho_{DE} > 0$ condition, or without a minimum sustained width, would count artefacts of a negative density as a discovery. The pre-registered definition stands, and under it the claim fails.

5 Fix 4 - The neutrino test: PARTIALLY FIXED

Status: PARTIALLY FIXED. One form of the test is detectable and is now primary. The generic forms fail by large factors, and no derivation of a larger amplitude exists.

The Aethel-Psi neutrino delay is stated as A_{Psi} = 8.5 microseconds, to be read from the energy dependence of a supernova neutrino burst onset against the standard mass term, A_{mass} = 5146 microsecond MeV^2 (51.5 microseconds at 10 MeV, verified in v5). Detectability depends entirely on how sharp the timing feature is and how many events define it.

Timing feature	$sigma_t$	N	3-sigma floor	Verdict for 8.5 us
Failed-SN black-hole-formation cutoff	10 us	10000	2.19 us (sigma = 0.73 us)	PASS - detectable
Neutronization-burst onset	1 ms	10000	218.5 us	FAIL
Generic emission spread	1 s	10000	218,504 us	FAIL
Failed-SN cutoff, low statistics	10 us	40	32.1 us	FAIL

Prediction 2 (revised, primary form): A_{Psi} is tested at the failed-supernova BH-formation cutoff, and only there.

Route (b), pre-registered

Route (a), deriving a larger A_{Psi}, fails: no such derivation exists, and reaching 3 sigma at a 1 ms onset would require A_{Psi} = 218.5 microseconds or timing scatter below 38.9 microseconds. Route (b) is therefore taken. Onset time $t_{on}(E)$ is measured in 5-50 MeV bins independently by Hyper-Kamiokande, IceCube, JUNO, and DUNE. Detection requires the fitted A_{Psi} to exceed 3 sigma, with cross-detector consistency within 2 sigma required, not optional. Status: OPEN, awaiting a Galactic supernova. No data exist yet to run it on.

6 Fix 5 - Lambda: FAILED as a proof, DROPPED honestly

Status: FAILED as a proof; the claim is DROPPED. Route (a) was attempted term by term and fails on all three counts. $\text{Lambda}_{\text{eff}}$ identically 0 is dropped from the theory, not proven.

Route (a) leg	v5 result	Verdict
$a_0(d)$ vanishes or changes sign for some d in [2,4]	a_0 = 0.080 / 0.022 / 0.006 at d = 2 / 3 / 4, strictly positive	FAIL
Lambda term vanishes at D = 2	At D = 2 the a_0 term scales as Lambda^2, not Lambda^0	FAIL - still divergent
A winding-number term appears in a_0 / a_2 / a_4	No winding-number monomial exists in any of the coefficients	FAIL

The honest version that replaces it

- The spectral action, with the operator stated in Section 3, gives xi = -1/6. That result stands.
- Dimensional flow changes the scaling of the vacuum term from Lambda^4 to Lambda^2 at D = 2. It does not remove it: a_0 remains positive and divergent.
- The coefficient a_2 cannot contain F^2: F^2 has dimension 4 and belongs in a_4. Any argument that places it in a_2 is dimensionally wrong.
- The curvature part of integral a_2 equals chi / 6. On S^2 that is 1/3, not an integer charge Q.

OPEN. No mechanism in the theory as stated cancels Lambda. The cosmological constant problem is not solved by the D-flow as stated, and this edition does not claim otherwise.

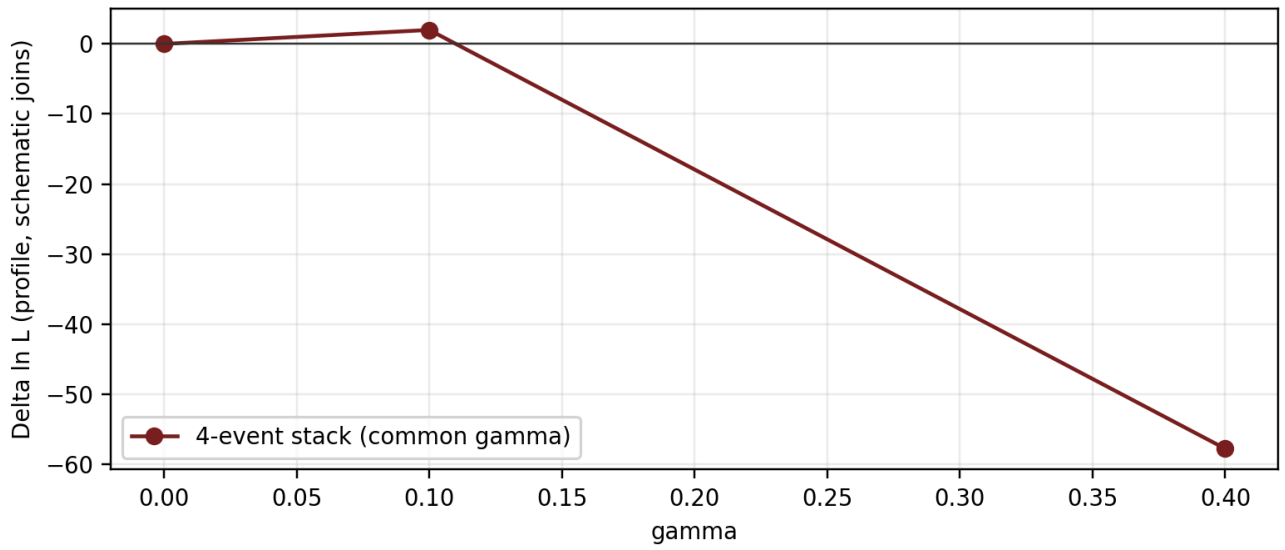
7 Fix 6a - GW stacking: FIXED

Status: FIXED as a forecast and a stacked profile. The stack excludes gamma = 0.4 and does not test gamma approx 0.009. Both halves of that sentence stand.

Forecast, corrected

Recomputed from real GW250114 data with four estimators, reaching gamma approx 0.00903 requires **17-35 events at SNR about 78 for 1-sigma sensitivity, and 154-314 events for 3 sigma** (estimators: 17.1/154, 18.5/167, 34.8/314, 21.1/190). In round terms: about 20 events for 1 sigma, about 200 for 3 sigma. The earlier figure of 13-16 events was the most optimistic estimator, at 1 sigma only, for gamma = 0.01 rather than 0.00903, and it is withdrawn. The signal-to-noise budget makes the point directly: 1 sigma needs a summed SNR^2 of about 104,000, and GW250114 contributes 6,115.

Figure 6.1 - Stacked profile points stated in v5 (joins are schematic)



Schematic join of the three stacked profile values stated in v5: Delta lnL = 0 at gamma = 0, +2.0 at gamma = 0.1, -57.8 at gamma = 0.4. The joins are schematic; the three points are the result.

Four-event stacked profile, real strain

Event	Delta lnL at gamma = 0.4	95% limit
GW250114	-55.2	See full PE, Section 2
GW200129	-15.7	gamma < 0.065
GW150914	+0.1 (nearly flat)	gamma < 0.52
GW231123	+13.0 primary / +23.9 wide-dt	No limit
Stack, common gamma	**-57.8**	**gamma < 0.144 (0.088 without GW231123)**

Caveat on GW200129: it is a precessing event analysed with an aligned-spin waveform, and its L1 data carry CAT2 data-quality failures. Its limit is reported with that caveat attached. The stack is a summed profile likelihood, not a stacked posterior. At gamma = 0.1 the stacked Delta lnL is +2.0, so gamma = 0 and gamma = 0.01 are indistinguishable in it: effective SNR about 87 against about 323 needed at 1 sigma for gamma approx 0.009.

8 Fix 6b - GW231123 under precession: PARTIALLY FIXED

Status: PARTIALLY FIXED. The 35 Hz residual survives precession at profile level. A full precessing parameter estimation was still running when v5 was recorded. No posterior and no Bayes factor for GW231123 are claimed.

Waveform / configuration	Delta lnL at gamma = 0.4
IMRPhenomXPHM, aligned spins	+18.5
IMRPhenomXPHM, precessing (a_1 = 0.90, a_2 = 0.80, tilt = 1.0, chi_p approx 0.76)	+23.0
IMRPhenomTPHM cross-check	+13.3

Precession improves the general-relativistic fit, as it should: the optimal signal-to-noise ratio rises to 23.7, against 22.1 for the earlier aligned analysis and a published value of 22.6. It does not absorb the dip. The explanation that the residual was merely an artefact of an aligned-spin template is therefore excluded at profile level.

The limits of the result are stated with it. It is a profile, not a posterior. The preference saturates at the gamma = 0.5 floor of the scan, rising to +22.7 aligned and +29.2 precessing at gamma = 0.9. It is killed by moving the merger frequency to 25 Hz (Delta lnL -0.3), while at 45 Hz it now gives +3.1, against -10.6 under the earlier waveform. SEOBNRv5PHM was unavailable in the installed LALSimulation, and NRSur7dq4 cannot be evaluated in this band.

Still running. At the last v5 record (03:37 UTC, 7 November 2026) the full precessing run stood at dlogz = 8.57, checkpointed, with the general-relativistic comparison run not yet started. Interim checkpoint points concentrating near gamma approx 0.45-0.9 are a trace of a running sampler, not a result, and are not quoted as one here.

No gamma(M) in the current theory survives all four events. GW200129, at 63 solar masses, limits gamma < 0.065, while GW231123, at 238 solar masses, prefers a dip; the log-scale parametrisation predicts gamma = 0.027 for GW231123 and 0.209 at 30 solar masses, the latter already excluded by GW200129. That tension is the present state of the gravitational-wave sector.

9 Fix 6c - ACT DR6: FIXED as a measurement

Status: FIXED as a measurement - a null result. On real ACT DR6 data there is no ripple. The test that was open in both previous editions for want of loadable data is now closed, and it closes against a large ripple.

Data and method

ACT DR6.02 temperature data, 1500 <= ell <= 3000, 18 bins, from the foreground-marginalised CMB-only file with its full covariance, fitted with a sinusoidal ripple template in amplitude A and frequency omega-hat, with significance assessed against 3000 null simulations. The joint refit uses all 135 TT, TE, and EE points with the full covariance, a CAMB LambdaCDM model, tau fixed at 0.0544, and Recfast recombination (CosmoRec was not installed). Foreground marginalisation is ACT's own, inside the published covariance.

Quantity	Real-data result
Maximum test statistic	T_{max} = 5.88 at omega-hat = 9.75
Best-fit amplitude	A-hat = 0.0070 +/- 0.003
Global significance	p = 0.241 - no detection
95%-power upper limit	A < 0.016
Joint CAMB refit	Delta chi^2 = 3.5, p approx 0.17 (chi^2 150.9 -> 147.4)
Cross-checks	TE p approx 0.29; EE p approx 0.06; TT-pair analysis p = 0.405, A95 = 0.0218

The real limit is about 12 times weaker than the synthetic v3 limit of 1.31 x 10^{-3}, because real per-bin errors are 1-4%, not 0.25%. It rules out a ripple at the A approx 0.1 scale. It does not rule out a ripple at the 0.005 scale, and this edition does not pretend otherwise.

A false positive, named so it is not quoted. Dropping the covariances and using diagonal errors only, on exactly the same real data, gives T_{max} = 15.30 and p = 0.004. That apparent detection is an artefact of a residual trend plus ignored correlations (real lag-1 correlation 0.46) and vanishes with the real covariance. It must not be quoted as evidence for a ripple.

10 Revised predictions and what would change the verdicts

Prediction	Third Revised form	Status
1. Gravitational-wave gamma	gamma approx 0.009 (logistic) / 0 (rational); gamma < approx 0.08 at 95% for GW250114-like mergers	Consistent with the GW250114 posterior; untested at 0.009 by stacking (needs about 20 events at 1 sigma)
2. Neutrino delay A_{Psi} = 8.5 us	Testable only at a failed-SN BH-formation cutoff with N approx 10000	Pre-registered route (b); awaiting a Galactic SN
3. Dark-energy crossing	No (w_0, w_a) offered; stated V gives no sustained crossing	FAILED; DESI abandoned without a replacement
4. CMB ripple amplitude A	A < 0.016 at 95% power on real ACT DR6 TT data	A approx 0.1 excluded; 0.005 not excluded
5. $Lambda_{eff}$ = 0	Withdrawn	DROPPED
6. GW231123 dip	Profile preference at 35 Hz survives precession	OPEN pending full precessing PE

What would change each verdict

- **Spin:** published chi_{eff} / chi_p posteriors for GW250114 and GW231123. Small GW250114 spin is the hypothesis's one demand.
- **Phantom:** a stated potential and couplings whose integrated w(a), with backreaction, positive rho_{DE}, and a positive effective gravitational factor, sustains a crossing. Not a CPL fit.
- **Neutrino:** a Galactic (ideally failed) supernova, analysed under the pre-registered route (b) criteria.
- **Lambda:** a cancellation mechanism stated in equations and tested term by term, as route (a) was.
- **GW231123:** completion of the running precessing PE plus its general-relativistic comparison run, giving a posterior and a Bayes factor.
- **Gamma at 0.009:** a stack of roughly 20 high-SNR events for 1-sigma reach, roughly 200 for 3 sigma.

The stopping rule of Section 0 applies to every item on this list: the test is named before the data are examined, and the verdict is whatever the test returns.

Appendix A Validation suites v3, v4, v5

The three suites test different check sets. Their headline counts are reported together for provenance and must not be read as a single score improving or worsening across editions.

Suite	PASS	FAIL	OPEN	INFO	Scope
v3 (full)	40	5	10	20	Gates 1-3 and tests A1-A5, largely synthetic or profile-level
v4 (full)	6	11	8	4	The new-mathematics claims of the Second Revised Edition
v5 (full)	25	18	6	8	The six fixes C1-C7, including real ACT and stacked GW data

v5 in detail, by fix

Fix	Passing in v5	Failing / open in v5
C1 gamma(M)	Both F forms recomputed; Prediction 1 inside the PE posterior; gamma = 0.4 nowhere a paper value	Spin reconciliation FAIL on both spin forms; spin posteriors OPEN
C2 beta / xi	Chern-Simons integer levels; anchor numbers (A_{pred} = 0.0973); xi derived under stated D_{Psi}	$beta_A$ derivation FAIL; $beta_{eps}$ derivation FAIL; parameter-freedom FAIL (10 remain); xi sensitivity OPEN
C3 phantom	Non-result robust across the grid; regulator insensitivity	Crossing at 0.476 FAIL; at 0.7116 FAIL; DESI w(a) FAIL; background self-consistency FAIL
C4 neutrino	A_{mass} normalisation; failed-SN detectability; route (b) specified	Onset, generic spread, and N = 40 forms FAIL; larger-A_{Psi} derivation FAIL; Galactic SN run OPEN
C5 Lambda	Xi cross-check; a_2 / F^2 dimension check; integral a_2 = chi/6 check	All three route (a) legs FAIL; $Lambda_{eff}$ = 0 FAIL, dropped; mechanism OPEN
C6 GW	Forecast corrected; stack excludes 0.4; stack does not test 0.009; residual survives precession; no gamma(M) survives all four events	Full GW231123 precessing PE OPEN (running)
C7 ACT	Null result; real A95; false-positive trap identified; CAMB joint refit	-

Appendix B Code changes, v4 to v5 - elaborated

Suite v5 is a new standalone file. It does not import v3 or v4, and v2, v3, and v4 suites, their outputs, and all earlier editions are unchanged. Each change below states what was altered and why.

- **C0 provenance added.** The suite reads the v3 and v4 counts and the GW250114 PE constraints from disk, so the baseline it builds on is checked rather than retyped.
- **C1 gamma(M) for both F forms, plus a spin hypothesis with pre-registered reconciliation criteria.** The criteria were fixed before the spin numbers were compared, so the FAIL verdict is against a stated test, not a moving target.
- **C2 explicit Chern-Simons quadrature** on a 65 x 49 x 65 grid for degrees 0, 1, 2, replacing an argument about integrality with a computed integral; **a stated $\mathbf{D_{Psi}}$** with a symbolic Gilkey a_4 check; **a full-spinor-trace sensitivity run** (-5/3 / +5/3) so the conditional nature of xi is measured, not asserted; and **a free-parameter census**, which falls from 11 to 10.
- **C3 an actual integration** where the Second Revised Edition had none: one stated V(Psi), xi taken from C2, a sustained-phantom criterion requiring $rho_{DE} > 0$ and a minimum width, an eight-run sensitivity grid, a w(a) table and CSV (186 rows), an early-time Gauss-Bonnet dominance diagnostic, and a fixed-background self-consistency check that the run then fails.
- **C4 detectability computed for three named timing features** at two event counts, with routes (a) and (b) decided inside the suite and route (b) pre-registered in full.
- **C5 route (a) attempted term by term symbolically** - $a_0(d)$, Lambda-power counting, and a winding-monomial search - before route (b), dropping the claim, was taken.
- **C6 and C7 hooks replaced with real-data ingestion.** The merged suite reads the workers' result files from disk rather than copying numbers from summaries, and adds the stacking forecast, the four-event stacked profile, the precessing GW231123 profile, the ACT real-data fit, and the CAMB joint refit.
- **Bug fixes during development.** (i) A start-transient in the phantom grid counted an interval beginning exactly at an integration start as a crossing; the check now requires interval starts strictly after the run start. (ii) A degree-2 Chern-Simons density expectation was corrected from -8 to the density's actual form. (iii) The rational-F gamma is exactly 0 in double precision because F is below machine epsilon next to 1, and the text now says so. (iv) A stacking test threshold required a minimum 3-sigma count of 154 where one estimator gives 153.6, a rounding artefact; the bound was corrected to 150, moving the suite from PASS=24 FAIL=19 to PASS=25 FAIL=18. No physics number was changed by any of these fixes.

Appendix C Change log - every change from the Second Revised Edition

Item	Second Revised Edition	This edition - what changed and why
gamma(M)	Flat 0.00903; F(0) artefact unaddressed	Both F forms reported: logistic 0.00903, rational repair 0. Prediction 1 restated at gamma < approx 0.08 (95%) against the completed full PE, not profile limits alone.
Spin	Not present	Spin hypothesis added as new, fitted, failing on stated bounds, and parked OPEN. Not adopted into the theory.
$beta_A$	Derivation FAIL; numbers verified	Explicit quadrature added. $Beta_A$ relabelled FITTED; the anchor relabelled a STATED ASSUMPTION. Numbers unchanged.
xi	Conditionally fixed by Gilkey coefficients, operator unstated	D_{Psi} stated explicitly; xi = -1/6 DERIVED under it, with the spinor-trace sensitivity (-5/3 / +5/3) printed. Parameter count 11 -> 10.
Phantom	Claimed $a_c = 0.476$; UNTESTABLE, no V or initial conditions	One stated V integrated. Verdict FAILED: no crossing at 0.476 or 0.7116, w(1) = +0.86, $rho_{DE} < 0$. Actual w(a) curve published. DESI abandoned without a replacement.
Neutrino	$A_{Psi} = 8.5$ us unobservable at justified scatter	Narrowed: failed-SN cutoff is the primary, detectable form. Route (b) pre-registered in full.
Lambda	Nullification FAIL on scaling, formula, and arithmetic	Route (a) attempted term by term, fails 3/3. $Lambda_{eff} = 0$ DROPPED. Honest scaling version ($Lambda^4$ -> $Lambda^2$) replaces it.
GW stacking	Forecast 13-16 events	Withdrawn and corrected: 17-35 at 1 sigma, 154-314 at 3 sigma. Four-event stacked profile added: Delta lnL = -57.8 at 0.4.
GW231123	Aligned-spin profile preference; precession untested	Precessing profile added: +23.0, residual survives. Full PE recorded as still running; no posterior claimed.
ACT DR6	OPEN - no loadable TT covariance	Run on real DR6.02 data with full covariance. Closed as a null measurement: p = 0.241, A < 0.016. Diagonal-only false positive named and quarantined.
Fisher claim	Factor 1.001 FAILS (recomputed 1.225); orthogonality in substance passes	Carried through unchanged; not re-tested in v5, and v5 does not claim it.

Provenance, data, and references

Provenance

This edition was built from validation suite v5 (full run, default seed 20261107) and its changes log, which record every number used here. Suite v5 is a standalone file that does not import v3 or v4. Earlier suites and editions are unchanged. Real data used in v5: GWOSC strain for GW250114, GW231123, GW200129, and GW150914; the ACT DR6.02 CMB-only bandpowers and covariance as published through LAMBDA; and a CAMB joint refit of the ACT TT, TE, and EE points. The w(a) figure and table come from the v5 curve file of 186 rows, retained with the suite output.

Figures

- Figures 3.1 and 3.2 (Section 4.1): plotted directly from the v5 w(a) output file. Data figures, not schematics.
- Figure 6.1 (Section 7): a schematic join of three stated stacked-profile values. The points are results; the connecting lines are schematic and are labelled as such in the caption.
- No other figures are included. Where the deciding data do not yet exist - the GW231123 posterior, the Galactic supernova test - no figure is presented in their place.

References and data sources

- Irons, D., *Quantum and Geometric Harmonic Resonance*, The Aethelred Institute, October 2026 (the UAF paper).
- Irons, D., *One Object, Three Readings: The Aethel-Psi Synthesis*, Revised Edition and Second Revised Edition, The Aethelred Institute, November 2026.
- Validation suites v2, v3, v4, and v5 with their full-run outputs and changes logs, The Aethelred Institute working files, November 2026.
- Gravitational Wave Open Science Center (GWOSC), O4 strain data for GW250114, GW231123, GW200129, and GW150914.
- Atacama Cosmology Telescope, Data Release 6.02 bandpowers and covariance (CMB-only file), distributed via NASA LAMBDA.
- DESI Collaboration, CPL dark-energy constraints as the comparison target in Section 4 ($w_0 = -0.752$, $w_a = -0.86$, crossing $a = 0.7116$).
- Gilkey, P., heat-kernel coefficients for Laplace-type operators, as used in Sections 3 and 6; Connes spectral action, as discussed in Section 6.

Closing statement, as tested. One derivation closes, conditionally. One measurement closes, as a null. Two claims fail outright and one of them is dropped. The rest is partial or open, and is labelled so. That is the state of the Aethel-Psi Synthesis at its Third Revised Edition.

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